

SIMATS ENGINEERING

ASSESSMENT TOOL 1: Experiments

COURSE NAME:	Cloud Computing and Big Data Analytics	COURSE CODE:	CSA15
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COURSE OUTCOME COVERED

CO3: *Implement cloud-based solutions using cloud storage, web APIs, and platform services for real-world applications.*
(BL3)

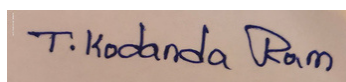
Dave's Taxonomy: Precision (Level 3) Question Count: 4	Total Marks: 40
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Code of Conduct

I T. Kodanda Ram (192524162) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: _____



Experiments Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Experimental Design	25%	Design is systematic and technically strong	Mostly correct design	Basic design with gaps	Poorly designed activity
Use of Tools / Platforms	20%	Appropriate and effective use of cloud tools	Good tool usage with minor issues	Limited tool usage	Incorrect or missing tool usage
Application of Concepts	20%	Concepts are applied accurately to practical tasks	Mostly accurate application	Partial application	Weak application
Analysis and Output	20%	Strong analysis and meaningful outcome interpretation	Good analysis with minor gaps	Basic analysis	Little or no analysis
Documentation	15%	Clear, structured, and complete documentation	Mostly complete documentation	Partially complete	Poor documentation

Total Marks Awarded:

Assessment Tasks

Experiments

Experiment 1: Create a simple cloud software application for Library book reservation system for SIMATS library using any Cloud Service Provider to demonstrate SaaS include the necessary fields such as Title of the book, author, publications, Year, edition, no of copies, Rack no, status (taken/returned) etc.

Experiment 2: Create a simple cloud software application for Payroll Processing System using any Cloud Service Provider to demonstrate SaaS include the necessary fields such as Name of the employee, experience in present org , over all experience, basic pay, HRA, DA, TA, net pay, gross pay etc.

Experiment 3: Demonstrate virtualization by Installing Type-2 Hypervisor in your device, create and configure VM image with a Host Operating system (Either Windows/Linux). Create a Virtual Machine with 1 CPU, 2GB RAM and 15GB storage disk using a Type 2 Virtualization Software.

Experiment 4: Demonstrate virtualization by Installing Type-2 Hypervisor in your device, create and configure VM image with a Host Operating system (Either Windows/Linux). Create a Cloning of a VM and Test it by loading the Previous Version/Cloned VM

Assessment Tasks

Experiments

Experiment 1: Create a simple cloud software application for Hospital information system for SIMATS Hospital using any Cloud Service Provider to demonstrate SaaS with fields of patient name, age, address, phone, email, Attender name, attender phone, doctor name, diseases, etc.

Experiment 2: Create a simple cloud software application for online super market using any Cloud Service Provider to demonstrate SaaS with the template of customer name, address, phone, email, items, quantity etc. with the item classification form.

Experiment 3: Demonstrate virtualization by Installing Type-2 Hypervisor in your device, create and configure VM image with a Host Operating system (Either Windows/Linux). Create a Virtual Machine with 3 CPU, 10GB RAM and 10GB storage disk using a Type 2 Virtualization Software.

Experiment 4: Demonstrate virtualization by Installing Type-2 Hypervisor in your device, create and configure VM image with a Host Operating system (Either Windows/Linux). Create a Snapshot of a VM and Test it by loading the Previous Version/Cloned VM.

Assessment Tasks

Experiment 1: Create a simple cloud software application for Mark Sheet Processing System using any Cloud Service Provider to demonstrate SaaS include the necessary fields such as name of the student, Reg no of the student , Subjects , Marks, Average, percentage, rank, overall performance etc.

Experiment 2: Create a simple cloud software application for Property Buying & Rental process (In Chennai city) using any Cloud Service Provider to demonstrate SaaS include the necessary fields such as Buy, Rent, commercial, Rental Agreement , Rental Agreement, Area, place, Range, property value etc.

Experiment 3: Demonstrate virtualization by Installing Type-2 Hypervisor in your device, create and configure VM image with a Host Operating system (Either Windows/Linux). Create a Virtual Machine with 5 CPU, 12 GB RAM and 8 GB storage disk using a Type 2 Virtualization Software.

Experiment 4: Demonstrate virtualization by Installing Type-2 Hypervisor in your device, create and configure VM image with a Host Operating system (Either Windows/Linux). Create a Snapshot of a VM and Test it by loading the Previous Version/Cloned VM.

C03-AT1 - High Experiments

① Hospital Information system using saas.

Aim: To create a simple cloud software application for hospital information system for SMATS hospital using a cloud service provider & demonstrate software as a service (saas)

Procedure: (i) open the zoho creator and login to zoho account.

(ii) create a new application and enter the application.

(iii) select one application.

(iv) Enter Application name.

(v) created new application

(vi) select one form.

(vii) The software has been created.

Implementation.

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SMATS HOSPITAL - PATIENT INFORMATION

Patient Registration Form

Patient name :

Age :

Address :

Phone :

Email :

Attender name :

Attender phone :

Doctor name :

Disease/Diagnosis :

Submit your details

output:

SRMATS Hospital - Patient Dataset					
ID	Patient name	Age	Phone	Doctor	Disease
1	Ram	20	9176844881	Dr. Yash	A cidity
2	Kaishna	22	7691448313	Dr. Kumar	piles
3	Koti	25	8844556781	Dr. Ravi	Fever
4	Ramesh	30	7977345468	Dr. mani	Cancer

Results Thus, the SRMATS Hospital Information system was successfully created using SaaS, stored in cloud online supermarket using SaaS.

② Aims To create a simple cloud software application for an online supermarket using a cloud service provider and demonstrate SaaS.

Procedures (i) select a cloud service provider such as Firebase

- (ii) create the online supermarket web application.
- (iii) Add customer, order, quantity, price and item classification fields.
- (iv) Enter sample customer and item details.
- (v) calculate the total amount and submit the order.

Implementation:

view customer order.

customer name

Kranti

order date

29-8-26

Address

VGP, near SMATS

payment method

online

phone

9988 776485

delivery option

Home delivery

output:

cloud dataset

SD	Name	Age	details	Item	quantity
1	D. Yashwanth	22	chennai	Rice	1
2	B. Kumar	21	madurai	milk	3
3	Chenna	24	Koyambedu	Fruits	1
4	Koushik	26	Selam	soaps	2

③ Hypervisor and virtual machine

Aim: To demonstrate virtualization by installing,
Type 2 - Hypervisor, creating and configuring
a virtual machine with host operating system,
and allocating 3 CPU - 10 GB RAM and
10 GB storage disk.

Procedures (i) download vmware workstation and ins as type 2 hypervisor.

(ii) download ubuntu or tiny as an iso file.

(iii) In vmware workstation → create new vm.

(iv) do the basic configuration settings.

(v) created tiny os virtual machine.

(vi) Launch the vm.

Implementation

my Linux 3cpu 10GB Ram
devices

memory 10GB

process, 3

Hard disk 10GB

CD/DVD : using file



(4) Snapshot of VM

Aim : To create a snapshot of a VM and test it by loading previous version / cloned VM.

Procedures (i) Goto vmware workstation.

(ii) create files on ubuntu.

(iii) click on vm & select snapshot.

(iv) snapshot is being done.

Implementation

my computed

other lines 2.2V

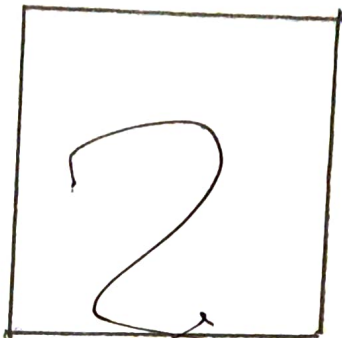
other lines 7.2V

other 2.2 x

send vms

Apps

search



onBoot TCB Tnplsnapshot

URL : <https://orpotis.ontes.pdf>